

DAVID CROPPER

AI Engineer

Melbourne, VIC | 0400 811 545 | dave_cropper2003@yahoo.co.uk

Portfolio: <https://journey.chaseriskandcompliance.com.au>

GitHub: <https://github.com/dcrops> | LinkedIn:

<https://www.linkedin.com/in/david-cropper/>

Professional Summary

AI Engineer with a background in Data Engineering and Automation Engineering. Experienced building governance-aware AI systems, Retrieval-Augmented Generation (RAG) platforms, workflow intelligence solutions and operational AI applications. Combines enterprise experience delivering production data platforms with hands-on development of FastAPI services, Streamlit applications, Dockerised solutions, retrieval systems, evaluation frameworks, telemetry and explainable AI workflows.

Core Technical Skills

Python • FastAPI • SQL • Docker • Streamlit • ChromaDB • OpenAI APIs • Git/GitHub • REST APIs • RAG • Retrieval Engineering • Evaluation Frameworks • Telemetry & Observability • Prompt Engineering • Databricks • AWS • Azure • Workflow Automation

Featured AI Engineering Projects

Governance-Aware Document Intelligence RAG Platform

- Designed and built a modular RAG platform for trustworthy document retrieval and explainable AI interactions.
- Implemented PDF, DOCX and TXT ingestion pipelines with configurable chunking strategies.
- Built retrieval orchestration, query routing, source attribution and grounding validation workflows.
- Developed FastAPI backend, Streamlit frontend and ChromaDB vector storage.
- Implemented evaluation frameworks, telemetry and observability for system performance analysis.
- Containerised and deployed using Docker.

Payroll Diagnostics & Governance Engine

- Designed a modular diagnostics platform for analysing payroll and operational datasets.
- Implemented ingestion, validation, diagnostics, findings and reporting layers.
- Developed rule-based analysis covering leave, termination, long service leave and integrity reviews.
- Generated evidence-backed findings, severity scoring and executive reporting outputs.
- Focused on explainability, traceability and governance-oriented decision support.

Public Holiday Entitlements Application

- Built an operational rules engine for Australian public holiday entitlement determination.
- Implemented geographic resolution, LGA matching and regional entitlement logic.
- Produced auditable outputs and compliance-focused reporting.

Professional Experience

Founder & AI Engineer — Chase Risk & Compliance (2025–Present)

- Founded a technology-focused consultancy used as a vehicle for building governance-aware AI and operational intelligence systems.
- Designed and built the Payroll Diagnostics Engine, Public Holiday Entitlements Application and Document Intelligence RAG Platform.
- Applied principles of explainability, observability, governance and trustworthy AI outputs.

Data Engineer — NBN Australia (2020–2023)

- Designed and developed scalable data pipelines supporting operational reporting and analytics.
- Worked extensively with Python, SQL, Databricks, APIs and Azure technologies.
- Implemented validation controls and production-grade data workflows.
- Partnered with technical and business stakeholders to deliver enterprise data solutions.

Senior Test Automation Engineer / QA Automation Engineer

- Developed enterprise automation frameworks and API testing solutions.
- Built regression, integration and end-to-end testing capabilities.
- Established strong foundations in automation, reliability and systems thinking.

Certifications

- AWS Certified Developer – Associate
- Databricks Data Engineer Associate
- Databricks Data Engineer Professional
- ISTQB Certified Tester
- Certified Agile Tester